BlinkMoney Save Journey

Data Analyst Intern Home Assessment — Final Write-up

Executive summary

The Save journey should be measured as a sequential funnel from app open to the first successful unit credit. The supplied profile shows the largest apparent loss between investment amount chosen and bank account verified: 2,290 versus 1,098 distinct users, an apparent loss of 1,192 users and approximately 48.0% conversion. This is directional until event order, duplicate rows, identity stitching, and asynchronous observation windows are validated.

I recommend 10 outcome events, with external confirmations separated from user actions. The north star is first unit credited within 30 days; the counter-metric is early SIP failure or cancellation. This aligns growth with genuine customer value.

Funnel snapshot

Stage	Users	Conv.	Lost
App opened	4,102	100.0%	—
Mobile verified	2,974	72.5%	1,128
Identity check passed	2,338	78.6%	636
Amount chosen	2,290	97.9%	48
Bank verified	1,098	48.0%	1,192
Mandate authorised	731	66.6%	367
Profile saved	688	94.1%	43
Consent confirmed	640	93.0%	48
SIP created	631	98.6%	9
First units credited	588	93.2%	43

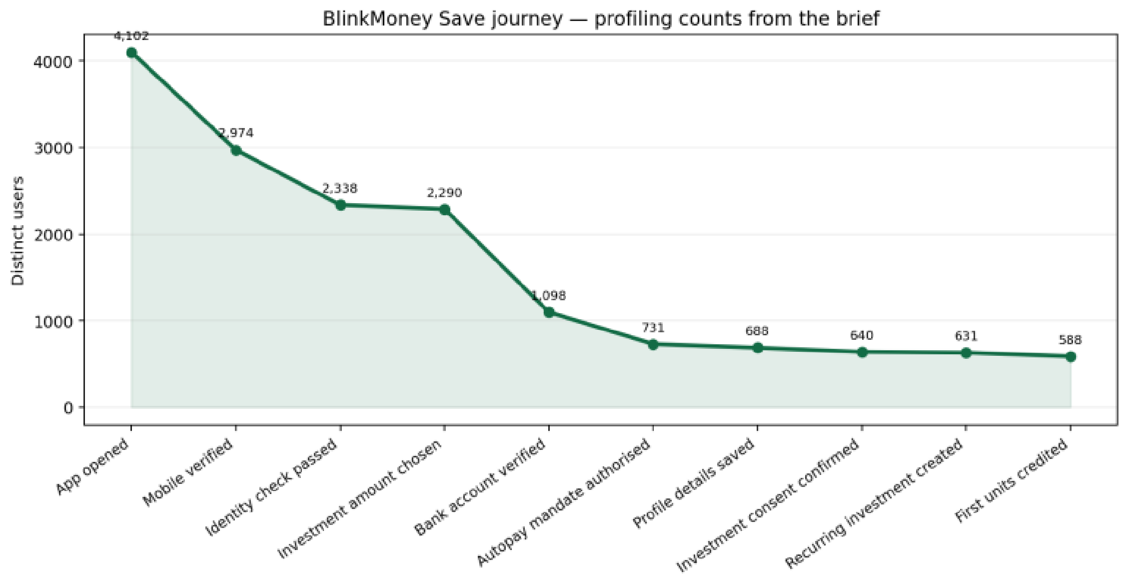

Figure 1. BlinkMoney Save journey — distinct-user counts per stage, plotted from the funnel snapshot above.

Insights and recommendations

- Bank verification is the first diagnostic priority: test provider failures, eligibility, abandonment, and trust/consent friction separately.
- Downstream creation is strong: consent-to-SIP creation is about 98.6%, so improving earlier stages may yield more value than redesigning final consent.
- Rows are not users: 14,205 amount-selection rows from 2,290 users imply retries or edits; user-level deduplication is essential.
- Recommended actions: instrument bank-link initiation and failure codes, add pending/recovery states for mandate and RTA credit, test a fallback bank provider, and build a mature-cohort dashboard with latency and reconciliation metrics.

Part A — Event specification

Naming: lowercase snake_case; outcome names are emitted only after the relevant success condition. Common fields are user_id, event_id, event_ts, app_version, platform, and session_id. Never send PAN, bank numbers, OTPs, or UPI PINs.

Event	Trigger	Properties (why)	Question / drop response
app_open	App becomes active after splash; lifecycle event.	platform, app_version, acquisition_source, session_id.	Are users entering the journey? Drop suggests launch/acquisition/technical friction; improve startup reliability.
mobile_verified	Successful API response after mobile OTP submission; confirmed.	attempt_count, retry_count, entry_point.	Can users verify mobile? Drop suggests OTP delivery/trust friction; improve retry and errors.
pan_verified	Successful PAN/KYC provider response or callback; confirmed/asynchronous.	provider, status, failure_code, latency_ms.	Who passes identity and what delays/fails? Drop suggests eligibility/provider/data mismatch; add recovery and status polling.
sip_amount_selected	Tap Continue after amount, frequency, scheme and allocation pass validation; user action.	amount_band, frequency, scheme_count, allocation_hash.	What configurations do users commit to? Drop suggests affordability/choice friction; simplify choices and defaults.
bank_verified	Bank verification API/webhook success after link and confirmation; confirmed.	provider, status, failure_code, latency_ms.	Can users link an eligible bank? Drop suggests coverage/provider/consent issues; add fallback and guidance.
mandate_completed	Lender/mandate-provider webhook confirms UPI authorisation; confirmed/asynchronous.	provider, status, failure_code, initiated_at, confirmed_at, latency_ms.	Do users obtain authorised autopay? Drop suggests UPI handoff/rejection/delay; improve deep link and pending status.
additional_details_saved	Server acknowledges required profile fields saved.	form_version, validation_error_count, nominee_selected.	Can users complete required profile? Drop suggests form burden; remove nonessential fields and improve validation.
investment_consent_verified	Server verifies consent OTP after submit; confirmed.	attempt_count, retry_count, consent_version, failure_code.	Can users complete final consent? Drop suggests OTP/copy/trust friction; improve copy and recovery.
sip_created	Order service returns durable success and order ID; confirmed.	order_id_hash, amount_band, frequency, status, failure_code.	Was a recurring investment created? Drop suggests backend/order validation; reconcile and provide retry.
first_unit_credited	RTA/ledger webhook or reconciled batch confirms units; confirmed/asynchronous.	order_id_hash, asset_type, credit_date, provider, latency_from_creation.	Did real investment value arrive? Drop suggests settlement/RTA/reconciliation problems; alert and reconcile.

Events cut

I did not create separate events for Welcome, Home, email OTP, every SIP setup screen, bank-link tap, UPI launch/return, nominee selection, or dashboard view. These are useful diagnostics but do not earn priority under the 12-event cap; they can be properties, screen logs, error logs, or later additions. Nominee is optional and is therefore a property.

Asynchronous design

PAN, bank verification, mandate approval, and first-unit credit fire from confirmed API responses, webhooks, or reconciliation jobs — not from taps or returning to the app. Store initiated_at and confirmed_at, use provider/order references for idempotency, and expose pending/failure states operationally.

Part B — Falsifiable hypothesis

Hypothesis: Amount chosen → bank verified leaks the most users because users have demonstrated investment intent, but bank linking introduces eligibility, privacy, provider, and handoff friction. The profile shows 1,192 fewer distinct users at bank verification. I would be wrong if an ordered cohort analysis over a complete maturity window shows a larger adjacent drop, or if bank initiation-to-success is high while another stage has the lowest conversion after retries and delayed callbacks are handled.

Part C — North star and counter-metric

The north-star metric is the weekly number of new users who receive their first successful unit credit within 30 days of first app_open. It measures delivered investment value rather than clicks or intent. The counter-metric is the 30-day early SIP failure/cancellation rate among credited users. I would report both by acquisition source, amount band, provider, and configuration. The north star is healthy only when the counter-metric does not materially worsen.

Part D — SQL write-up

The attached SQL file maps the 10 events to ordered stages, deduplicates to the first timestamp per user/event, and uses a recursive CTE to enforce that each stage occurs after the previous stage. It then returns a labelled funnel output, median elapsed seconds from app_open to first_unit_credited for users completing all stages, and the largest absolute adjacent-stage loss.

Expected SQL interpretation

mandate_completed is confirmed by the lender or mandate provider; first_unit_credited is confirmed by the RTA or asset ledger. Their timestamps represent operational completion, not user taps. Recent cohorts may therefore look artificially incomplete unless the reporting period includes the expected settlement/credit window.

Assumptions

- Event names exactly match the stage map.
- user_id remains stable across sessions, devices, and provider callbacks.
- Repeated events are possible; the SQL keeps the first timestamp for each user and event.
- A sequential funnel counts a later stage only when it occurs after the prior stage.
- First stage conversion is 100%; later conversion is users reaching current stage divided by previous stage.
- Completed means all 10 stages are observed; median time is elapsed seconds from first app_open to first unit credit.
- The credit observation window is mature; otherwise pending users must be separated from failures.
- All timestamps use a consistent timezone and trusted clock source.
- Test/internal users, duplicate callbacks, and event retries are excluded or deduplicated.

- If users can create multiple SIPs, a journey_id/order_id is required to avoid mixing journeys.
- The supplied distinct-user profile is not assumed to be sequential; it must be reconciled with the SQL result.

Data checks before trusting results

Check nulls, duplicate event_ids, duplicate provider references, timestamp timezone/clock skew, event-name drift, stable identity stitching, test users, extraction completeness, callback delays, impossible sequences, multi-SIP users, and provider/RTA reconciliation. Compare raw rows with distinct users and manually trace a sample of user journeys against order and investment records.